

William Zhang

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EDUCATION

Purdue University (John Martinson Honors College)

West Lafayette, IN

Honors B.S. in Computer Science and Mathematics; GPA: 3.93/4.00

Expected May 2028

Coursework: Data Structures & Algorithms, Computer Architecture, Probability and Statistics, Programming in C, Linear Algebra, Real Analysis, Ordinary Differential Equations, Data Mining and Machine Learning, Web Information Search and Management, Systems Programming

EXPERIENCE

Machine Learning Intern

May 2026 – Aug 2026

Stesso AI

Portland, OR

- Cut video-generation API token use by 16% while maintaining output quality, reducing inference costs; built an OpenCV/PyTorch repair classifier for home-inspection photos with 83% test accuracy on Kaggle data.
- Built a search pipeline for 1,663 instructional-video clips across 250 videos; evaluated TF-IDF, structured matching of action, object, tool, and supply fields, and hybrid ranking with blinded relevance tests.

Mobile App Developer

May 2026 – Present

The Peer Network (Telehealth Startup)

Remote

- Developed and launched a React Native telehealth app for iOS and Android on an eight-person engineering and design team; served 900+ users, increased appointment rebooking by 34%, and published on the App Store and Google Play.
- Integrated Amazon Chime for HIPAA- and SOC 2-compliant sessions between licensed peer counselors and patients; implemented a Cronofy booking system with mutex synchronization to prevent double-booking.

Project Lead and Full-Stack Developer

Sep 2025 – Present

Purdue Stack

West Lafayette, IN

- Led development of an Electron.js interface for SeisGo, cutting seismic-analysis time by 34% compared with the lab's Jupyter workflow.
- Added Fourier-series analysis to SeisGo, an open-source Python package on PyPI.

RESEARCH

Undergraduate Researcher

Aug 2026 – Present

Purdue University Department of Computer Science – Professor Sotiris Nousias

West Lafayette, IN

- Built a Python and NumPy simulator and browser interface for single-photon detection, modeling photon arrivals, efficiency, dark counts, dead time, jitter, and picosecond timestamp quantization.
- Implemented Fourier analysis on photon timestamps; validated photon statistics and spectral peaks across sparse signals, aliasing, detector saturation, and pulse trains with numerical tests.

Research Intern

Jun 2024 – Aug 2025

University of Pennsylvania Perelman School of Medicine

Philadelphia, PA

- Analyzed 500,000+ emergency room cases spanning 20 years in Professor Chamith Rajapakse's lab; trained a PyTorch/Keras network to predict bone fracture risk from environmental and movement features.
- Built an Arduino/C++ mobility tracker for physical therapy assessment; co-authored a poster accepted for presentation at the 2025 New England ACSM conference.

PROJECTS

Finmath — PyPI — Python, C++, pybind11, NumPy

Jan 2026 – Present

- Built and published Finmath, a C++-backed Python package on PyPI, with Purdue's Boiler Quant club.
- Implemented C++ time-series tools and Black-Scholes and binomial-tree pricing with pybind11 Python bindings.

Ecofriend — React Native, JavaScript, TypeScript, Java, Swift, Expo

May 2024 – Present

- Published a disaster-recovery app for the 2024 Congressional App Challenge on the App Store and Google Play.

TECHNICAL SKILLS & AWARDS

Languages: Java, Python, C, C++, JavaScript, TypeScript, R, SQL

Frameworks/Tools: React Native, Node.js, Next.js, Flask, Electron, Git, GitHub, Docker, AWS, Linux

ML/Data: PyTorch, scikit-learn, Keras, OpenCV, NumPy, pandas, SciPy, Matplotlib

Awards: USA Computing Olympiad (Gold Division, 2024 US Open Contest)